Ziyi Francis Yin, Ph.D.

Email: ziyiyin97@outlook.com — Last update: Jun 2026
Website: ziyiyin97.github.io — Based in: California, United States
Google scholar: https://scholar.google.com/citations?user=ji9kwj8AAAAJ
Research interests: inverse problems, generative models, simulation-based inference and imaging

WORK EXPERIENCE

KronosAI — California, US
Founding Member of Technical Staff — Feb 2025 - Jun 2026
- AI-accelerated physics simulation and inverse design

Occidental Petroleum Corporation — California, US
Research Geophysicist — Aug 2024 - Jan 2025
- Generative AI and Scientific ML workflows for uncertainty-aware geophysical imaging and inversion

Chevron Corporation — Houston, TX
Research Intern — May 2023 - Aug 2023
- Scalable and cloud-native differentiable programming frameworks for multiphysics inversion

EDUCATION

Georgia Institute of Technology — Atlanta, GA
Doctor of Philosophy in Computational Science and Engineering — Aug 2024
Master of Science in Computational Science and Engineering — May 2023
Advisor: Felix J. Herrmann — GPA: 3.85/4

Emory University — Atlanta, GA
Bachelor of Science in Applied Mathematics — May 2019
Advisor: James G. Nagy — GPA: 3.98/4

JOURNAL ARTICLES

J10. Abhinav Prakash Gahlot, Rafael Orozco, **Ziyi Yin**, Grant Bruer, and Felix J. Herrmann. "An uncertainty-aware Digital Shadow for underground multimodal CO2 storage monitoring". Jul 2025. In: *Geophysical Journal International*. DOI: 10.1093/gji/ggaf176.

J9. Tamas Nemeth, Kurt Nihei, Alex Loddoch, Anusha Sekar, Ken Bube, John Washbourne, Luke Decker, Sam Kaplan, Chunling Wu, Andrey Shabelansky, Milad Bader, Ovidiu Cristea, and **Ziyi Yin**. "Superstep wavefield propagation". Apr 2025. In: *Wave Motion*. DOI: 10.1016/j.wavemoti.2024.103489.

J8. **Ziyi Yin**, Rafael Orozco, and Felix J. Herrmann. "WISER: Multimodal variational inference for full-waveform inversion without dimensionality reduction". Mar 2025. In: *Geophysics*. DOI: 10.1190/geo2024-0483.1. Featured in Geophysics Bright Spot in The Leading Edge.

J7. **Ziyi Yin**, Mathias Louboutin, Olav Møyner, and Felix J. Herrmann. "Time-lapse full-waveform permeability inversion: A feasibility study". Aug 2024. In: *The Leading Edge*. DOI: 10.1190/tle43080544.1.

J6. **Ziyi Yin***, Rafael Orozco*, Mathias Louboutin, and Felix J. Herrmann. "WISE: Full-waveform variational inference via subsurface extensions". Jul 2024. In: *Geophysics*. DOI: 10.1190/geo2023-0744.1. Honorable Mention for Best Paper in Geophysics in 2024. Featured in Geophysics Bright Spot in The Leading Edge.

* denotes equal contribution.

J5. **Ziyi Yin**, Rafael Orozco, Mathias Louboutin, and Felix J. Herrmann. "Solving multiphysics-based inverse problems with learned surrogates and constraints". Oct 2023. In: *Advanced Modeling and Simulation in Engineering Sciences*. DOI: 10.1186/s40323-023-00252-0.

J4. Thomas J. Grady II, Rishi Khan, Mathias Louboutin, **Ziyi Yin**, Philipp A. Witte, Ranveer Chandra, Russell J. Hewett, and Felix J. Herrmann. "Model-parallel Fourier neural operators as learned surrogates for large-scale parametric PDEs". Sep 2023. In: *Computers & Geosciences*. DOI: 10.1016/j.cageo.2023.105402.

J3. Mathias Louboutin*, **Ziyi Yin***, Rafael Orozco, Thomas J. Grady II, Ali Siahkoohi, Gabrio Rizzuti, Philipp A. Witte, Olav Møyner, Gerard J. Gorman, and Felix J. Herrmann. "Learned multiphysics inversion with differentiable programming and machine learning". Jul 2023. In: *The Leading Edge*. DOI: 10.1190/tle42070474.1. Featured in Seismic Soundoff. Journal's most downloaded paper in 2023.

J2. Yijun Zhang, **Ziyi Yin**, Oscar Lopez, Ali Siahkoohi, Mathias Louboutin, Rajiv Kumar, and Felix J. Herrmann. "Optimized time-lapse acquisition design via spectral gap ratio minimization". Jul 2023. In: *Geophysics*. DOI: 10.1190/geo2023-0024.1.

J1. **Ziyi Yin**, Huseyin Tuna Erdinc, Abhinav Prakash Gahlot, Mathias Louboutin, and Felix J. Herrmann. "Derisking geologic carbon storage from high-resolution time-lapse seismic to explainable leakage detection". Jan 2023. In: *The Leading Edge*. DOI: 10.1190/tle42010069.1.

PEER-REVIEWED CONFERENCE PROCEEDINGS

C9. Tamas Nemeth, Kurt Nihei, Alexander Loddoch, Anusha Sekar, Ken Bube, John Washbourne, Luke Decker, Sam Kaplan, Chunling Wu, Andrey Shabelansky, Ovidiu Cristea, and **Ziyi Yin**. "Finite-difference wavefield propagation using superstepping". Jul 2024. In: *Fourth International Meeting for Applied Geoscience & Energy Expanded Abstracts*. DOI: 10.1190/image2024-4091674.1.

C8. Abhinav Prakash Gahlot, Huseyin Tuna Erdinc, Rafael Orozco, **Ziyi Yin**, and Felix J. Herrmann. "Inference of CO2 flow patterns – a feasibility study". Oct 2023. In: *NeurIPS 2023 Workshop - Tackling Climate Change with Machine Learning*. DOI: 10.48550/arXiv.2311.00290.

C7. Yijun Zhang*, **Ziyi Yin***, Oscar Lopez, Ali Siahkoohi, Mathias Louboutin, and Felix J. Herrmann. "3D seismic survey design by maximizing the spectral gap". Aug 2023. In: *Third International Meeting for Applied Geoscience & Energy Expanded Abstracts*. DOI: 10.1190/image2023-3895546.1.

C6. Huseyin Tuna Erdinc*, Abhinav Prakash Gahlot*, **Ziyi Yin**, Mathias Louboutin, and Felix J. Herrmann. "De-risking Carbon Capture and Sequestration with Explainable CO2 Leakage Detection in Time-lapse Seismic Monitoring Images". Nov 2022. In: *AAAI 2022 Fall Symposium - The Role of AI in Responding to Climate Challenges*. DOI: 10.48550/arXiv.2212.08596.

C5. **Ziyi Yin**, Ali Siahkoohi, Mathias Louboutin, and Felix J. Herrmann. "Learned coupled inversion for carbon sequestration monitoring and forecasting with Fourier neural operators". Aug 2022. In: *Second International Meeting for Applied Geoscience & Energy Expanded Abstracts*. DOI: 10.1190/image2022-3722848.1. Student Oral Paper Honorable Mention.

C4. Mathias Louboutin, Philipp A. Witte, Ali Siahkoohi, Gabrio Rizzuti, **Ziyi Yin**, Rafael Orozco, and Felix J. Herrmann. "Accelerating innovation with software abstractions for scalable computational geophysics". Aug 2022. In: *Second International Meeting for Applied Geoscience & Energy Expanded Abstracts*. DOI: 10.1190/image2022-3750561.1.

C3. Yijun Zhang, Mathias Louboutin, Ali Siahkoohi, **Ziyi Yin**, Rajiv Kumar and Felix J. Herrmann. "A simulation-free seismic survey design by maximizing the spectral gap". Aug 2022. In: *Second International Meeting for Applied Geoscience & Energy Expanded Abstracts*. DOI: 10.1190/image2022-3751690.1.

C2. **Ziyi Yin**, Mathias Louboutin, and Felix J. Herrmann. "Compressive time-lapse seismic monitoring of carbon storage and sequestration with the joint recovery model". Sep 2021. In: *First International Meeting for Applied Geoscience & Energy Expanded Abstracts*. DOI: 10.1190/segam2021-3569087.1.

C1. **Ziyi Yin**, Rafael Orozco, Philipp A. Witte, Mathias Louboutin, Gabrio Rizzuti, and Felix J. Herrmann. "Extended source imaging – a unifying framework for seismic and medical imaging". Sep 2020. In: *SEG Technical Program Expanded Abstracts 2020*. DOI: 10.1190/segam2020-3426999.1.

THESES

T2. **Ziyi Yin**. "Solving geophysical inverse problems with scientific machine learning". Aug 2024. *PhD dissertation*. URL: https://hdl.handle.net/1853/75649.

T1. **Ziyi Yin**. "Edge Detection and Enriched Subspaces". May 2019. *BSc Dissertation*. URL: https://etd.library.emory.edu/concern/etds/7w62f916x.

PROFESSIONAL SERVICE

Editorial Service
Geosciences
- Topical Advisory Panel Member
- Guest Editor: Special Issue on Geophysical Inversion

Technical Program Committee
AAAI 2023 Fall symposium on Artificial Intelligence and Climate

Session Chair
International Meeting for Applied Geoscience and Energy 2023, 2024

Journal Reviewer
Acta Geophysica, Algorithms, Applied Sciences, Axioms, Computers, Computers and Geosciences, Current Research in Geoscience, Earth Sciences, Earth Science Informatics, Electronics, Energies, Fractal and Fractional, Future Internet, Geophysics, Geophysical Prospecting, Geoscientific Model Development, IEEE Transactions on Geoscience and Remote Sensing, Interpretation, Journal of Applied Geophysics, Journal of Geophysics and Engineering, Journal of Geophysical Research: Machine Learning and Computation, Journal of Geophysical Research: Solid Earth, Journal of Mathematics and Statistics, Journal of Open Research Software, Journal of Open Source Software, Mathematics, Pure and Applied Geophysics, Processes, Remote Sensing, Scientific Reports, Sensors, Sustainability, The Leading Edge

Conference Proceeding Reviewer
AAAI 2023 Fall symposium on Artificial Intelligence and Climate
ICLR 2024 workshop on AI4DifferentialEquations
International Meeting for Applied Geoscience and Energy 2023, 2024
Neurips 2024 workshop on Data-driven and Differentiable Simulations, Surrogates, and Solvers
SciMLCon 2022
58th US Rock Mechanics / Geomechanics Symposium

Award Reviewer
Georgia Tech President's Undergraduate Research Award 2022, 2023

ACADEMIC SERVICE

Georgia Institute of Technology Geophysical Society — Atlanta, GA
President — Oct 2020 - Sep 2022
Secretary — Nov 2019 - Oct 2020

Emory University Office of Undergraduate Studies — Atlanta, GA
Academic Fellow — Aug 2018 - May 2019

TEACHING EXPERIENCE

Georgia Institute of Technology Atlanta, GA
Teaching Assistant, Seismic Monitoring CO_2 Storage Spring 2022
Head Teaching Assistant, Computational Data Analysis Fall 2021
Teaching Assistant, Exploration Seismology Spring 2021
Teaching Assistant, Iterative Methods for Systems of Equations Fall 2020

Emory University Atlanta, GA
Teaching Assistant, Probability and Statistics I & II Fall 2018, Spring 2019
Teaching Assistant, Foundation of Mathematics Summer & Fall 2018, Spring 2019

HONORS AND AWARDS

2024 Honorable Mention for Best Paper in Geophysics
2022 IMAGE's Student Oral Paper Honorable Mention
2021 SEG Technical Program Registration grant
2020 SEG/Chevron Student Leadership Symposium travel grant
Graduate with Highest Honors (*summa cum laude*), Emory University May 2019
Phi Beta Kappa Honor Society Membership Apr 2019
Dean's List, Emory University Aug 2017 - May 2019

GRANTS

SEG Field Camp grant ($1000) 2022
Studying 1886 Earthquake at Summerville, South Carolina – Seismic Nodal Deployment in the Field

SKILLS

Languages: Julia, Python, MATLAB, Java, C/C++, Bash, SQL, PHP, R, MPI
Machine Learning Libraries: PyTorch, Tensorflow, JAX, Flux.jl
Web & Front-End: React.js, Flask, Vercel
Cluster/Cloud Service Platforms: Slurm, Amazon Web Services (AWS), Microsoft Azure, Docker
Document Preparation Systems: Markdown, LaTeX, html